

6	C	For the bob, $T \cos 11^\circ = mg \text{ and } T \sin 11^\circ = ma$ $\Rightarrow \tan 11^\circ = a / g$ $\Rightarrow a = g \tan 11^\circ = 1.90687 \text{ m s}^{-2}$ For the block and bob, $\Sigma F = (\Sigma m) a$ $250 - F_f = 100 \times 1.90687$ $F_f = 59.3 \text{ N}$
7	C	Area under $F-t$ graph gives the change in momentum of the body